

Project 22: Movie Revenue Prediction

Midterm Report

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1 Progress and Implementation Details

At this midterm stage, we have successfully moved from the preliminary design to the active implementation phase of our movie revenue prediction model. Following the task guidelines, the objective is to predict movie revenue from a set of features, which we frame as a continuous numerical regression problem.

We are utilizing the official TMDb Box Office Prediction dataset from Kaggle. Our current training pipeline operates on an extracted subset of 3,000 complete records, examining the following attributes:

- **budget** (integer): The financial budget allocated for the film.
- **runtime** (float): The total duration of the movie in minutes (contains 2 missing values which are dynamically handled).
- **original_language** (string/categorical): The primary language in which the movie was filmed.
- **revenue** (integer): The global box office revenue, serving as our continuous target variable.

To ensure strict compliance with the project’s reproducibility requirements, we have initialized a global random seed (`SEED = 42`). This seed governs our data splitting strategy—maintaining an 80% training set and a 20% validation set—and fixes the stochastic initialization of our ensemble models.

The core pipeline is built using `scikit-learn`’s robust workflows. Continuous features (`budget`, `runtime`) are automatically processed via a pipeline that handles missing values through median imputation and normalizes scale variance using a `StandardScaler`. Categorical features (`original_language`) undergo missing value imputation followed by a sparse `OneHotEncoder` transformation.

2 Intermediate Results

We have completed the baseline model implementation stage, successfully training both a regularized linear model and a non-linear ensemble tree model. The performance was evaluated using Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score) on the unseen 20% test partition.

The empirical results are summarized in Table 1 below.

Table 1: Preliminary Model Performance Metrics

Model Architecture	RMSE (\$)	R^2 Score
Ridge Regression (Baseline)	85,486,050.06	0.5655
Random Forest Regressor	94,943,898.48	0.4641

To visually inspect the prediction quality, the true box office revenue was plotted against the model predictions for both techniques. The resulting visualization is embedded below in Figure 1.

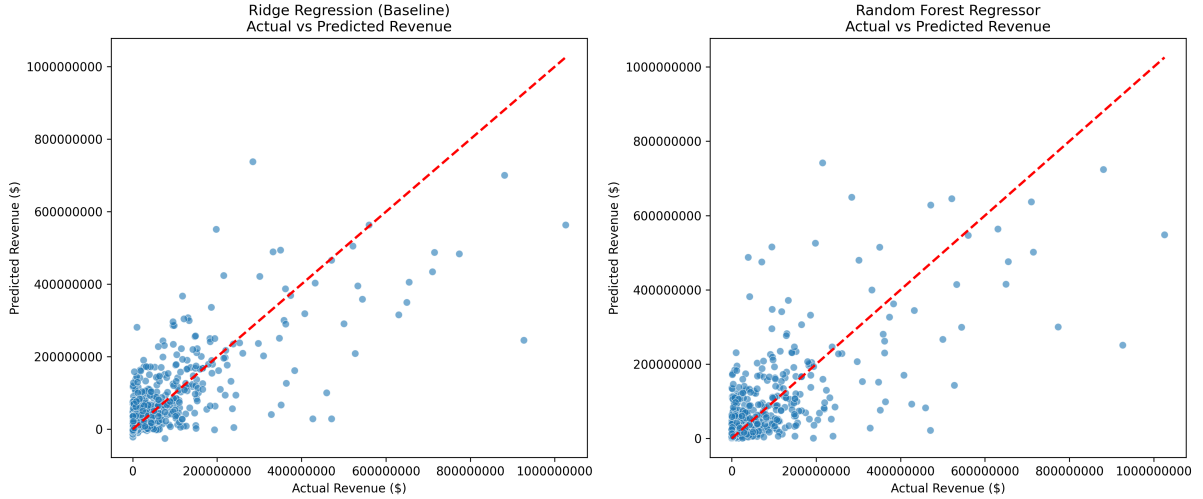


Figure 1: Actual vs. Predicted Revenue Scatter Plots for Ridge and Random Forest Models.

3 Challenges, Findings, and Corrections

The midterm evaluation revealed critical structural behaviors that required us to rethink our initial assumptions:

- **Baseline Outperformance Paradox:** An unexpected finding was that our baseline Ridge Regression model outperformed the more complex Random Forest Regressor (R^2 of 0.5655 vs. 0.4641). Analysis of the error distribution shows that the target variable (**revenue**) is highly right-skewed with extreme out-of-distribution blockbusters. Untuned Random Forests cannot extrapolate beyond the maximum target value seen in their training leaf nodes, leading to severe underestimations of high-revenue films. Ridge regression, due to its linear extrapolation nature, handled these heavy tails better.
- **Missing Data Imputation:** The `runtime` column exhibited missing elements (2 non-null rows missing out of 3,000). Dropping rows would shrink our dataset, so we corrected this by integrating a `SimpleImputer` set to the feature’s median, stabilizing training.
- **High-Cardinality Categorical Features:** High variance in the `original_language` column creates sparse matrices during One-Hot Encoding. For the final phase, we plan to group low-frequency languages into a unified “Other” token to prevent overfitting in the linear baseline.

4 Plans for Finishing the Project

To finalize the project and achieve competitive predictive accuracy, our remaining milestones are structured as follows:

1. **Advanced Gradient Boosting Implementation:** We will implement our third proposed model, the `XGBoost Regressor`. Since gradient boosted trees optimize residuals sequentially, it is expected to handle non-linear structures better than an untuned Random Forest while minimizing residual variances.
2. **Hyperparameter Optimization:** We will conduct a rigorous hyperparameter sweep using `GridSearchCV` or `RandomizedSearchCV`. For Random Forest, we will tune `max_depth`, `min_samples_split`, and `n_estimators`. For XGBoost, we will optimize the learning rate (`eta`) and tree regularization constants.
3. **Target Variable Transformation:** To resolve the skewness challenge identified in Section 3, we will test training our models on a log-transformed target space, $\log(\text{revenue} + 1)$, which will regularize the loss function against extreme blockbuster leverage.
4. **Ablation Study and Feature Sourcing:** As mandated by the project criteria, we will perform an ablation study to isolate the predictive value of text metadata (e.g., movie titles or taglines) versus financial indicators, and investigate external data augmentation like inflation scaling.